

A One-Dimensional Model of the Gravitational Assist Slingshot Effect: An Experimental Study

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November 2024 - January 2025

Abstract

This project investigates the gravitational assist slingshot effect through a simplified one-dimensional model, focusing on energy transfer during collisions between a tennis ball (the "spaceship") and a football (the "planet"). Using a computer-aided tracking algorithm, we precisely measured the velocity and maximum rebound height of both objects during the experiment. A pneumatically-driven testing rig, equipped with adjustable tripods, ensured precise control of the drop height and minimized human error, and the tennis ball was 3D-printed to achieve a consistent weight and bounce. Experimental results showed that, as predicted by our theoretical model, the tennis ball rebounded to a height greater than its initial drop height, confirming the energy transfer predicted in the gravitational assist process. Our experimental results align with our theoretical model, validating the simplified theory used in this study.

1. Introduction

Gravitational assist, also known as a gravity slingshot, is a technique used by spacecraft to gain speed and adjust their trajectory by passing close to a planet or moon. This technique relies on the conservation of momentum and energy to transfer kinetic energy from a planet to the spacecraft, enabling the spacecraft to achieve higher velocities without expending additional fuel. The concept was first introduced by Yuri Kondratyuk in 1938, in his paper "To Those Who Will Be Reading in Order to Build" ("Тем, кто будет читать, чтобы строить")^[1]. In this work, he proposed that a spacecraft travelling between two planets could use the gravity of the moons of those planets to gain additional speed at both the beginning and the end of its journey. This concept was later refined by Michael Minovitch in 1961. Minovitch's work was pivotal in demonstrating that by flying a spacecraft through the gravitational field of a planet, the spacecraft's velocity could be increased without the need for fuel, thus making long-duration missions feasible. This led to his

technique later being used in Gary Flandro's Planetary Grand Tour idea^{[2][3]}.

At its core, gravitational assist operates under the principles of classical mechanics, specifically the conservation of momentum and energy. When a spacecraft passes near a planet, its trajectory is altered due to the planet's gravitational field. The spacecraft gains kinetic energy as it is accelerated by the planet's gravity, while the planet's motion is minimally affected due to its much greater mass. This interaction can be described using the following equation for velocity change in a gravitational field:

$$\Delta v = \frac{2GM}{r} \cdot \frac{1}{v}$$

where G is the gravitational constant, M is the mass of the planet, r is the distance from the planet's center, and v is the velocity of the spacecraft. This equation represents the basic dynamics that govern the slingshot effect, illustrating how the relative motion between the spacecraft and the planet results in a change in the spacecraft's velocity.

Our experimental model draws from these fundamental concepts of energy transfer during collisions. We examine the interaction between a football and a tennis ball in a simplified system, where the football represents a "planet" and the tennis ball represents a "spaceship." By analyzing the velocity changes and energy transfer during elastic collisions between the two objects, we aim to model how a spacecraft might experience velocity changes during a gravitational assist manoeuvre. The experimental setup, involving a pneumatically-driven rig, is designed to minimize human error and ensure precise measurements of velocity and height changes.

The goal of this study is to compare the results of the experiment with the theoretical predictions derived from the principles of momentum and energy conservation, thus producing a clear correlation between predictions and results. By doing so, we can explore how energy transfer occurs in gravitationally assisted collisions on a small scale, offering insights into the mechanisms and factors behind this crucial technique.

2. Theoretical Model

2.1 Background Theory

To begin to understand the theory of gravitational assists, we must abstract it down to its most applicable parts. As mentioned previously, this effect can be modelled through the principles of conservation of momentum and kinetic energy as in a perfectly elastic collision, both are conserved. Consider our experiment; a system consisting of a football with mass m_f , and a tennis ball with mass m_s , where $m_F \gg m_T$. These masses will remain constant. Let v_1 denoted velocity of the system before rebound, and v_2 denote velocity after the collision. When they are dropped together from variable height h (as measured from the centre of the football), the velocity of the football, v_1 (as $v_{1F} = v_{1T}$), upon impact with the floor, can be modelled as:

$$v_1^2 = u^2 + 2gh \quad (1)$$

where g is acceleration due to gravity. $u = 0$, (1) so this can be rearranged to give:

$$v_1 = \sqrt{2gh} \quad (2)$$

As the ground is significantly more massive than the football, it will rebound with same the magnitude, in the opposite direction. This can be expressed as:

$$v_2 = -\sqrt{2gh}$$

The football then collides with the tennis ball on rebound, which, in this instance, is still moving down with velocity v_1 . We can now model this as a simple 1-dimensional elastic collision where particle 1 (football) is much more massive than particle 2 (tennis ball), where their velocity is equal but in opposite directions. Let v_{2T} and v_{2F} represent the balls' velocities after the collision. Then, using the same assumptions, we can use the general elastic collision velocity equation, $v_{\text{after}} = \frac{(m_1 - m_2)v_1 + 2m_2v_2}{m_1 + m_2}$ to give the velocity of the tennis ball after rebound as:

$$v_{2T} = \frac{(m_T - m_F)v_{1T} + 2m_Fv_F}{m_T + m_F} \quad (3)$$

We can now substitute $v_T = \sqrt{2gh}$, as it is still moving downwards, and $v_F = -\sqrt{2gh}$, as the football has already rebounded off the floor, into (3) to give:

$$v_{2T} = \frac{(m_T - m_F)(\sqrt{2gh}) + 2m_F(-\sqrt{2gh})}{m_T + m_F} \quad (4)$$

$$v_{2T} = \frac{m_T\sqrt{2gh} - m_F\sqrt{2gh} - 2m_F\sqrt{2gh}}{m_T + m_F} \quad (5)$$

$$v_{2T} = \frac{m_T\sqrt{2gh} - 3m_F\sqrt{2gh}}{m_T + m_F} \quad (6)$$

Factoring out $\sqrt{2gh}$:

$$v_{2T} = (\sqrt{2gh}) \frac{m_T - 3m_F}{m_T + m_F} \quad (7)$$

as $m_F \gg m_T$, we can use a high-mass approximation, as in the real model, the planet of m_F would have a much greater mass than the spaceship of m_T to the point where m_T is practically insignificant. Therefore:

$$\frac{m_T - 3m_F}{m_T + m_F} \approx \frac{-3m_F}{m_F} \approx -3$$

Substituting $\sqrt{2gh}$ back into this high-mass approximation gives:

$$\Rightarrow v_{2T} = -3\sqrt{2gh} \quad (8)$$

using our original kinematic equation (1), this can be rearranged to give the rebound height as $h_T = \frac{(v_{2T})^2}{2g}$. Substituting (8) into this will give us the theoretical rebound height with a perfect system to be:

$$h_T = \frac{(-3\sqrt{2gh})^2}{2g} \Rightarrow \frac{9(2gh)}{2g} \Rightarrow h_T = 9h$$

Using this theoretical value of the rebound height, we can compare this to our experimental findings found in 4. to comment on the accuracy of our model.

2.2 MATLAB Simulation of the Model

Before conducting any physical experiments, we first developed a computer simulation in MATLAB, based on the theory outlined in 2.1. This simulation allowed us to visualise the expected outcomes, and provided a basis for comparing results.

We are able to vary both the height of the drop and the coefficient of restitution (a measure of how elastic a collision is) in our simulation. The coefficient of restitution, e , represents the ratio of the relative velocity of the tennis ball after impact, to its velocity before impact. For simplicity, we set $e = 0.75$, which is a reasonable - even expected - value for a tennis ball^[6]. This is slightly lower than the theoretical maximum of 1.0, due to real-world energy losses such as heat and sound.

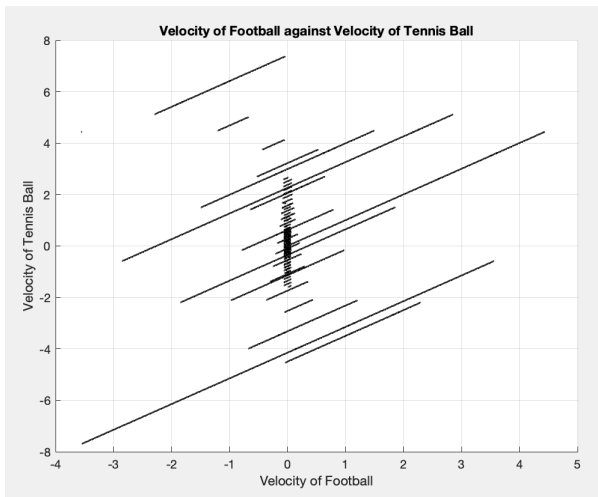


Figure 1: Scatter graph of max v_T , against v_F $m = 3.01789$

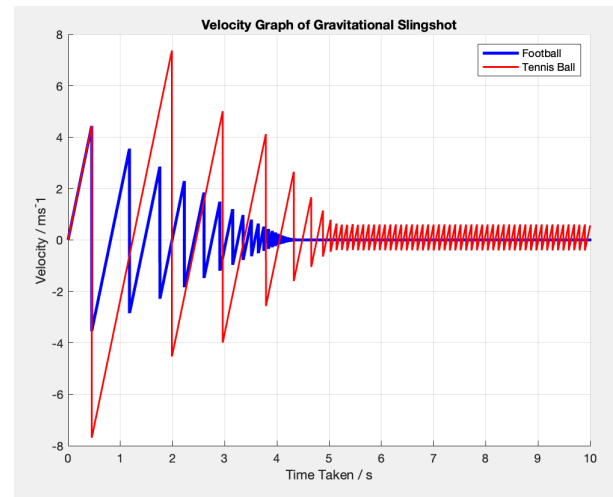


Figure 4: Graph of velocities, v_T , against time, t

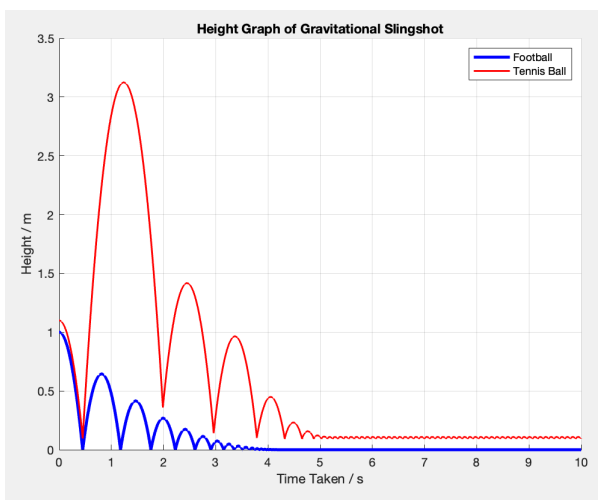


Figure 2: Graph of both ball heights, h_T , against time, t

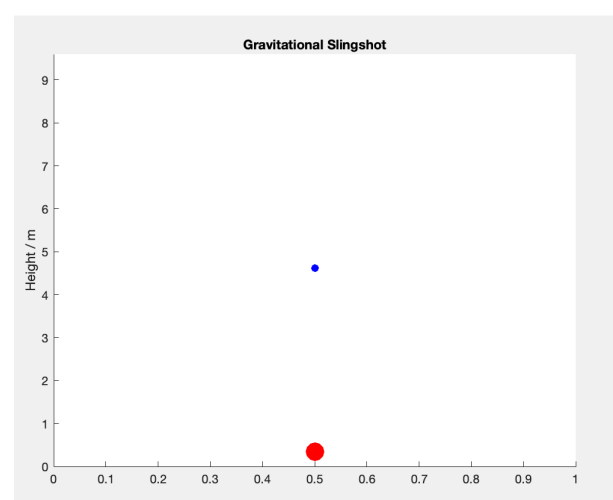


Figure 5: Visualisation of the MATLAB simulation

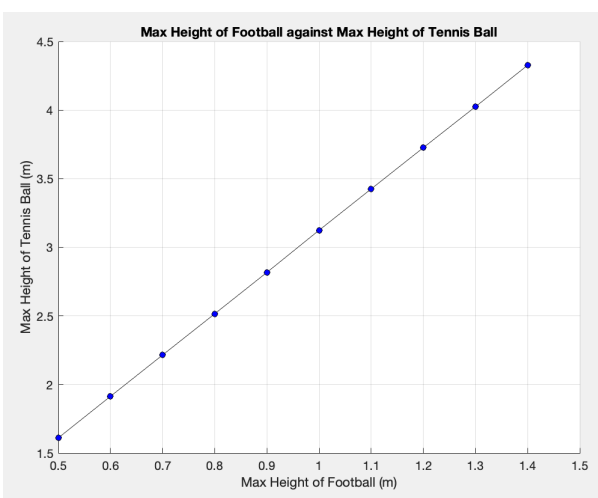


Figure 3: Scatter graph of max tennis ball height, h_T , against max football height, h_F $m = 3.01789$

Our MATLAB code simulated the motion of the two balls under gravity, updating their heights and velocities at each timestep, using equations like the one shown below. For the full program, refer to the dataset in **Section 7**.

$$v2Final = ((m2 - m1) / (m1 + m2)) * v2 + ((2 * m1) / (m1 + m2)) * v1;$$

The graphs presented here show the results of the simulation, including the height and velocity of both the football and the tennis ball over time. The main focus of our simulation is on **Figure 3**, which will be compared with the experimental value later. This shows the linear relationship between the maximum rebound heights, helping us hypothesise what the results of our experiment should look like.

3. Experimental Model

3.1 Establishing the Experimental Procedures

Before beginning any experiments, the experimental procedure and technique had to be established. This began with the selection of balls to be used for testing.

3.1.1 Determining Ball Specifications

Initially, four commercially available balls were purchased, including two **Size 5** footballs - as this is the standard size used in all adult competitions^[7]. The choice of a **Size 5** football was made to ensure that the balls would be closest in diameter to a typical "football", thus staying as true as possible to the original experiment. Additionally, a **Size 2** football was purchased, which is the smallest competition-legal football^[7]. This was done in case further tests are required, such as different mass or size specifications. The tennis ball selected was from a well-known manufacturer, HEAD^[8], ensuring it adhered to the International Tennis Federation's guidelines for competition-legal tennis balls, with a diameter range of 6.54–6.86 cm^[9].

Before taking any measurements, an electrical air pump was used to inflate the footballs to the ranges specified by the manufacturers, as indicated on the balls themselves. This step ensured that the diameter measurements would be as accurate as possible, closely reflecting the true diameter of the footballs.

Instead of relying on the advertised size range for a **Size 5** football^[7], the diameter was empirically determined and measured. A horizontal beam was mounted to a universal stand, and it was slid down until it made contact with the top of each ball. The height from the floor to the bottom of the beam was then measured with a metre stick. All measurements were taken at eye level to avoid parallax error, and the metre stick's resolution was in millimeters, resulting in a measurement error of ± 0.5 mm. The diameter of the tennis ball was measured using digital calipers (brand: *Magnusson*), by moving the lower jaws inwards until they touched the outer surface of the ball. The resolution of the calipers was 0.01 mm, so the error was ± 0.005 mm.

The mass of each ball was measured with a laboratory digital mass balance. This had a resolution of 0.01 g, resulting in a measurement error of ± 0.005 g.



Figure 6: Purchased Footballs (left to right): *Euros 2024*, *Sondico CF Size 5*, *Sondico CF Size 2*

3.1.2 Selecting Balls for the Experiment

At this point, we needed to select the ball that would provide the most consistent performance. Our primary criteria were:

- A consistent rebound height
- A consistent landing location for bounces

Initial rough tests were conducted to determine which ball met these criteria most effectively. Each football was dropped from a height of 1.0 m, and the height to which it rebounded was measured using a metre stick. The test was repeated ten times for each ball, and the standard deviation, σ_h , of each ball's return height was calculated. The balls were dropped by hand, and the return height measured visually, as this was only a rough comparison.

We also marked a cross on the ground and dropped the balls directly onto it. The location of the second bounce was then measured by determining the x and y offsets from the cross. The absolute distance from the cross was calculated using the Pythagorean theorem, and the geometric mean, μ_v , of these measurements was also computed. These two values were used to assess the bounce consistency of each ball:

Euros 2024 Football:	$\sigma_h = 2.55$	$\mu_v = 4.06$
Sondico Crossfire Football:	$\sigma_h = 4.06$	$\mu_v = 8.59$

Ball Name	Adv. Size	Diameter, d /mm	Mass, m /g	Internal Air Pressure, p /psi
Euros 2024 Football	Size 5	215 ± 0.5	432.83 ± 0.005	10.4
Sondico Crossfire Football	Size 5	220 ± 0.5	433.09 ± 0.005	10
Sondico Crossfire Football	Size 2	146 ± 0.5	166.53 ± 0.005	6.5
HEAD Championship Tennis Ball	-	66.48 ± 0.005	58.94 ± 0.005	-

Table 1: Specifications of the purchased, commercially available balls.

The Euros 2024 football had the lower σ_h for the return height, at 2.55, and a mean horizontal offset that was closer to zero ($\mu_v = 4.06$), indicating that its bounce was more consistent than that of the Sondico Crossfire football. **Based on this data, we chose to use the Euros ball for the actual experiment.**

We speculate that the Sondico ball's poorer performance was due to its construction, which involves separate artificial leather panels stitched together. The seams between the panels caused irregularities in the ball's diameter, increasing the variation in its bounce. In contrast, the Euros ball is made from a single outer shell of multiple layers of polyurethane (PU) that are thermally bonded^[10], leading to a more consistent surface.

3.1.3 Manufacturing a Custom Tennis Ball

A similar test was performed for the tennis ball, where the return height and landing position were measured. As we only had one ball to test, we set a threshold for the maximum permissible standard deviation for both the return height ($\sigma_h \leq 3.00$), and for the mean horizontal bounce ($\mu_v \leq 5.00$).

Below are the results of this test (10 repeats):

HEAD Tennis Ball: $\sigma_h = 6.87$ $\mu_v = 8.12$

The standard deviation for both measurements exceeded the threshold we set. We suspect that the irregular diameter of the tennis ball caused this inconsistency, similar to the issue with the Sondico football. To improve consistency, we decided to manufacture a more uniform tennis ball with a consistent diameter. This would increase the overall reliability and precision of our experiment's measurements. To create a more consistent tennis ball, we designed a 3D model of a perfect sphere with the same diameter as the original tennis ball (66.48 mm). The ball was 3D printed using 95A TPU filament^[11], which allowed us to have more control over the material properties, such as shore hardness, Young's modulus, and tensile strength, by referring to the material datasheet^[12]. However, as the density of TPU is 1.22 g/cm³^[12], we could not print a solid ball without significantly increasing its mass compared to the original tennis ball. We also experimented with different infill types to determine which one would provide the most consistent bounce. Small test hemispheres were 3D printed, changing only the infill type. We tested each hemisphere by measuring its diameter, mass, shore hardness, and compression consistency. The compression consistency was assessed qualitatively by manually applying a force of 20.0 N $\pm$ 0.5 N with a digital Newton meter, and visually inspecting the deformation. The deformation was classified into three categories: Low, Medium, and High, based on its regularity.

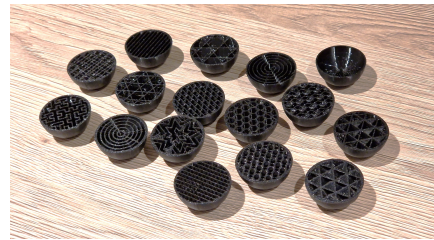


Figure 7: Infill Type Test Hemispheres

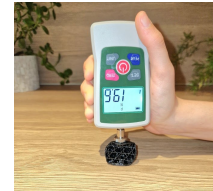


Figure 8: Ball Compression Test

After evaluating the results, we chose the Gyroid infill pattern for the final 3D printed tennis ball. This pattern provided a high deformation consistency, the diameter was closest to that of the original model (only a 0.19 mm difference), and the shore hardness was within the acceptable range. The final tennis balls were 3D printed in red 95A TPU^[11] to improve visibility, which would help in both visual recognition and any potential image recognition software used during testing. The key 3D print settings used for the final print are as follows:

- Layer height = 0.2 mm
- Line width = 0.42 mm
- Internal solid infill speed = 250 mm/s
- Wall loops = 7
- Top surface pattern = Concentric
- Top shell layers = 5
- Bottom surface pattern = Monotonic Line
- Internal solid infill pattern = Gyroid

The final tennis ball was printed using a Bambu Lab X1 Carbon 3D printer, and subsequent measurements showed a lower standard deviation in bounce height, improving the consistency compared to the original tennis ball. In addition to the ball, a 3D model of an O-ring was designed to place between the balls. This O-ring was modeled as a hollow cylinder with an outer diameter of 32.60 mm, inner diameter of 25.00 mm, and thickness of 3.80 mm. It was 3D printed using black PLA filament^[13].



Figure 9: 3D Printed Custom Tennis Ball

3.1.4 Consistent Ball Release

To ensure consistent drop heights, and eliminate the safety concerns, we developed an automated ball release mechanism. Initially, we explored a motorized trapdoor mechanism, wherein balls would be dropped from a container. However, this approach presented multiple issues: the motor's actuation speed was insufficient, the rebounding ball could strike the container, and the rolling motion off the trapdoor would introduce unwanted rotational forces. To overcome these challenges, we designed a claw mechanism, which secures the ball from the side and releases it via pneumatic actuation. This mechanism was designed using Autodesk Fusion, where it was modeled to be mounted on adjustable tripods, allowing for easy modification of drop height. **The adjustable frame can extend from 90 cm to 140 cm above the ground.** To incorporate heights of below 90cm, we increased the floor level directly below the dropping point using controlled MDF flooring.

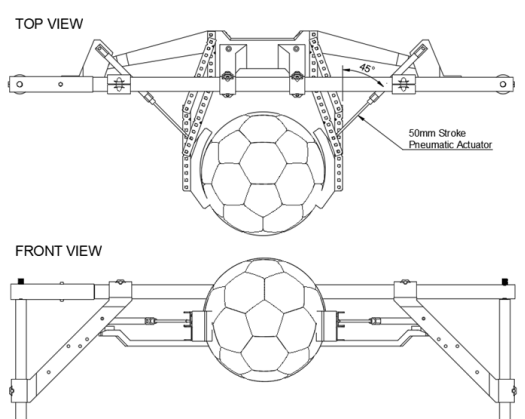


Figure 10: Orthographic Projection of Automated Ball Release Mechanism

A pneumatic system was used to control the release mechanism, with airflow through the cylinders managed via a microcontroller. The solenoid was controlled by a transistor to regulate the system's actuation.

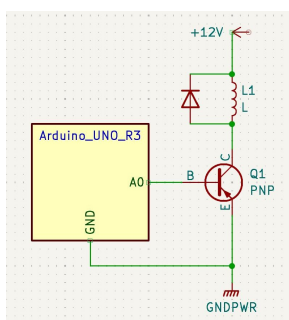


Figure 11: Schematic Diagram of Solenoid Control Logic

Most structural components of the release mechanism were 3D-printed and secured with 8-32 COTS screws and Nylock nuts. Extra care was taken to ensure the rig could reliably support the balls' weight.

3.2 Methods

3.2.1 Pre-Setup

We will control the orientation of the ball during dropings using the rotational tape and maintain the same perpendicular height of the dropping mechanism for each recording using a spirit level. To ensure consistency, a black stage curtain was set up as a backdrop, with the automated ball release mechanism positioned in front. A 15mm thick MDF sheet was placed directly below, to serve as a consistent drop surface, and an area, approximately 2.0m in radius, was marked and cleared of obstructions. The balls were illuminated from the sides using lightboxes, making them easily distinguishable against the black background.

A Panasonic Lumix S5 camera was placed 4.0m away from the setup, mounted on a tripod, with the focal length adjusted to 45mm to minimize lens distortion. The camera's framerate and resolution were set to 180 fps at Full HD (FHD) quality, to capture high-speed footage of the ball drops. To track rotational movement, tape lines were applied to each rotational axis of the football. A tape measure was mounted vertically, with green tape markers placed at 50cm intervals. This provided a clear reference for any post-experiment video calibration.

3.2.2 Procedure

1. Set release mechanism height, and ensure both sides are level, using a 600mm spirit level.
2. Pressurize the pneumatic air tank to ($90 < x \leq 100$ psi).
3. Place the football into the claw mechanism, ensuring correct orientation, with the tape lines perpendicular to the floor. Position the O-ring and the custom tennis ball on top of the football, centering them using the green tape lines.
4. H & S checks: Ensure safety goggles are being worn, inspect the equipment for electrical and pneumatic faults, and ensure all people stand outside the marked area.
5. A verbal countdown of "3, 2, 1, Drop" is given: at "2", begin camera recording, and on "Drop," release the ball using the remote trigger.
6. Wait for all balls to come to a complete stop, then stop the recording. Log the current time and recording number in the logbook.
7. Perform five repeats for each height, increasing the drop height by 10cm after each set.

Drop heights are restricted by, h :

$$h = \{x \mid 50 \leq x \leq 140, x = 10k, k \in \mathbb{Z}\}$$

3.3 Risk Assessment

To ensure the safety of our procedure, we conducted the following risk assessment before beginning to collect any data. This serves as a qualitative measure of the **Impact** of each of the risks, as well as outlining the mitigation strategy used, to eliminate or minimise each risk.

P = Probability (the likelihood that the hazard will occur, from 1 (low) to 5 (high))

S = Severity (how severe the consequence would be, from 1 (low) to 5 (high))

R = Overall Risk (calculated as $P \times S$, ranging from 1 to 25)

Impact : A qualitative measure of the effect, depending on the overall risk, and how easily it can be mitigated

Risk	P	S	R	Impact	Mitigation Used
Damage to surroundings	3	2	6	Low	Carry out experiment in an empty physics lab and a spare office room, with limited valuable possessions nearby.
Ball ricocheting sideways	4	4	16	High	Wear safety goggles throughout the experiment, and ensure the area is clear during drops.
Injury from ball drops	3	3	9	Medium	Mark out a 2.0m radius around the drop area, and stand behind this while the balls are in motion. Using a remote trigger for the ball drop.
Electrical shock from equipment	1	2	2	Low	Ensure all electrical connections are secure and inspected before use.
Data loss due to technical malfunction	1	5	5	Low	Save the video footage to two separate SD cards. Manually check at regular intervals that the recordings are being saved correctly.
Pressurised system failure	4	5	20	High	Ensure all pneumatic tubing is secure before pressurising. Only pressurise to 100psi.
Ball deformation affecting results	2	2	4	Low	Inspect balls regularly. Prepare substitute balls to use if required.

Table 2: Risk assessment table with P (Probability), S (Severity), R (Overall Risk), Impact, and Mitigation Used.

4. Experimental Findings

4.1 Results

Our experiment generated 50 recordings across 10 heights (see 3.2.2 for details).

4.1.1 Data Extraction from Recordings

To extract data from the videos, we developed a Python script using OpenCV. This script identifies the bounding boxes of the red (custom tennis ball) and white (football) balls within each frame as x and y coordinates (for details of this software, please access the link found in 7, file name *image_processor/image_processor.py*). The largest continuous red region is used to detect the red ball, and the green tape lines identify the football (Figure 11). We used the 50cm green interval markers to convert pixel values to millimeters, allowing us to calculate the height of both balls for each frame. With approximately 500 frames per video, we obtained around 25,000 data points in total.

The raw data had to be filtered and further processed. There is a delay between the start of recording and the actual ball drop, resulting in frames where the balls remain stationary. We wrote a script to remove this initial stationary segment

(*data_post_processor/data_filterer.py*). Additionally, we computed the velocity at each timestep using height measurements from consecutive frames with the equation $v(t) = \frac{dh}{dt}$, implemented in a separate python script (*data_post_processor/data_processor.py*). These programs were applied to all the videos, producing 50 .csv files containing height and velocity data for both balls in every frame (*data_post_processor/processed_results*).

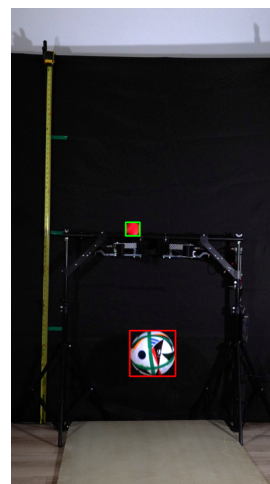


Figure 12: Image Tracking Software

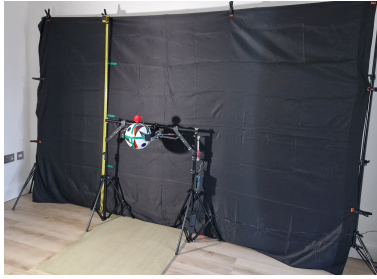


Figure 13: Complete Experiment Setup

4.1.1 Error and Accuracy Estimates

The resolution of the tape measure used to place the interval markers is $0.001\text{m} = 1\text{mm}$. This results in an uncertainty of $\pm 0.5\text{mm}$ in the positioning of these tape lines, corresponding to a relative uncertainty of 50%. The distance between the top and bottom tape lines is 764 pixels, providing a pixel-to-millimeter conversion

factor of 0.00131mm/px . Additionally, the detection accuracy of the ball in pixels carries an uncertainty of ± 1 pixel (relative uncertainty of 100%).

Combining these, the total relative uncertainty becomes

$$100\% + 50\% = 150\%$$

leading to an absolute uncertainty of

$$\pm 1.5\text{mm}$$

in the measurement of the balls' position.

It is likely that additional factors increase the uncertainty in our measurement, such as the accuracy of the image recognition model we are using, however there is no way to easily determine the uncertainty contributed by this.

4.2 Comparison to the Theoretical Model

Given that we had a lot of data, only a sample will be analysed here. To access the full data, please visit the GitHub Repository URL found at 7. The data expressed below is taken from dropping height of 1.0m (our median height value). Below is the processed data from our experiment aligned with the MATLAB simulation data for our system:

Table 3: Figure 1.1: Experimental data from height 1.0m

timestamp.s	red_relative_Y_mm	white_relative_Y_mm	red_velocity_mm.s	white_velocity_mm.s
0.0000	966.4751	827.2251	NaN	NaN
0.8174	212.5484	62.8272	1576.9880	1025.0422
2.0687	1199.4594	353.4031	78.3772	-470.2635

Table 4: Figure 1.2: MATLAB Simulated data from height 1.0m

Max Height (mm)	Max Rebound Velocity (mm/s)
3123.9	7693.1

Table 5: Figure 1.3: Maximum Theoretical Heights from 2.1

Drop height, h (mm)	Rebound height, $9h$ (mm)	Max Velocity, $v_T = \sqrt{2gh}$ (mm)
1000	9000	4429.44

Comparing the data above, we can deduce the differences in each of our models for this experiment. Using the theoretical model outlined in 2.1, the tennis ball should rebound to $9h$. We can deduce a negative correlation between realism of the model and rebound height with the MATLAB simulation data in 1.2. Here, using: $\frac{3123.9}{1000}$ is $\approx 3.1h$, a decrease of $\approx 65.3\%$. In figure 1.1, the rebound height is stated to be ≈ 1199.46 , using: $\frac{1199.46}{1000}$ is $\approx 1.2h$, a decrease of $\approx 86.7\%$. This is a significant decrease in the rebound height, which backs up our negative correlation hypothesis.

Table 6: Figure 1.4 Correlation Visualizer

Model	Rebound Height (mm)	Rebound multiplier (h)	Percentage of Theoretical Value ($9h$)
Theoretical	9000	$9h$	100%
Simulated	3123.9	$3.1h$	34.7%
Experimental	1199.46	$1.2h$	13.3%

This table clearly outlines our hypothesis above. This is due to more factors of the collision causing not all momentum and kinetic energy to be transferred into the tennis ball as outlined in 2.1, so less energy rebounded.

4.3 Conclusions

The primary aim of this experiment was to investigate how varying the drop height of the tennis ball and football would change the maximum height reached by the tennis ball on its return, due to the gravitational slingshot effect.

We chose to compare our experimental data to the graphs produced through our MATLAB simulations (2.2), in order to verify the accuracy of our test procedure and equipment. When plotting v_F and v_T against time, both the simulation and experimental results produced similar graphs. This is also true for the graphs of v_T against v_F , and h_T and H_F over time (see Figure 15). The consistency also suggests that the experiment followed the gravitational slingshot effect, as demonstrated in the MATLAB simulation.

From our data logs, we determined the maximum return height of the tennis ball and the maximum release height of the football for each trial. After compiling these values into a results table, we calculated the geometric mean of all the repeats, along with an uncertainty measure (see *other_data/max_ball_heights.csv* in the dataset). We did not identify any significant outliers, so we used all five data points when determining the mean.

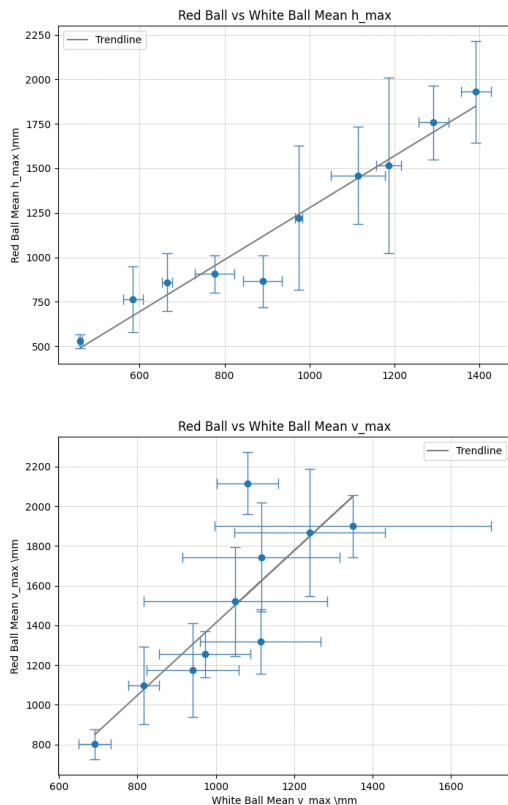


Figure 14: Scatter graphs plotting $h_{T \text{ MAX}}$ against $h_{F \text{ MAX}}$, and $v_{T \text{ MAX}}$ against $v_{F \text{ MAX}}$ (with uncertainty)

Linear Trendline: $y = 1.4601x - 181.9110$

R-Squared: $R^2 = 0.9535$

This data was then used to produce a graph of h_F against h_T (see Figure 14), which, upon visual inspection, showed a clear linear correlation. A linear regression line was fitted to the data with the equation $y = 1.4601x - 181.9110$. This linear relationship is expected, as a linear equation was derived in Section 2.1. The high R-squared value of 0.9535 further supports this correlation.

Upon comparison with theoretical model, we notice that the actual maximum height achieved by the tennis ball is roughly $\frac{1}{3}$ of that in the simulation. This can be explained by the inclusion of other forces, such as air resistance and energy losses during collisions, which were not fully accounted for in the theoretical model. The tennis ball's lower height in reality likely results from imperfect energy transfer between the two balls during the collision (the coefficient of restitution, $e < 1$), as well as frictional losses with the ground.

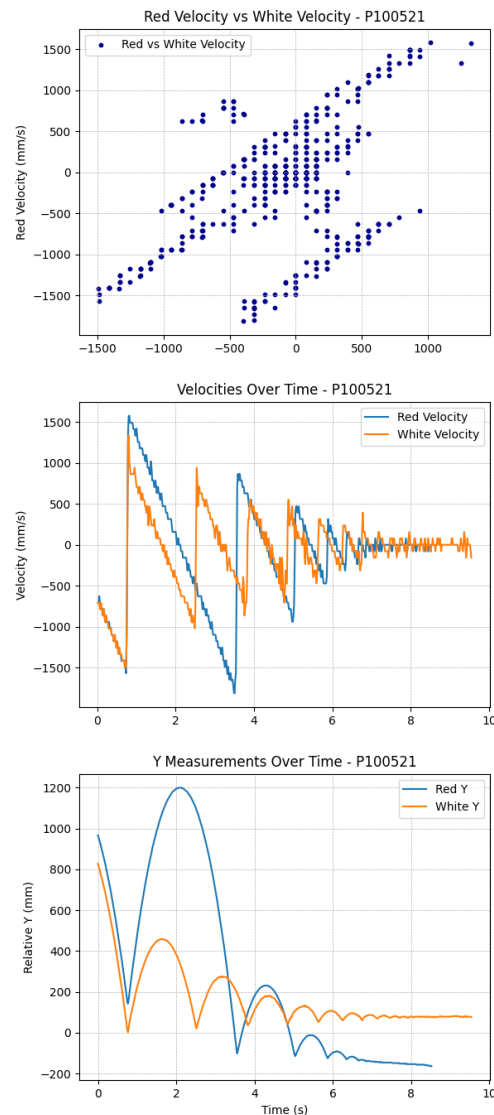


Figure 15: Graphs for P100521 (Repeat 3 at height 100cm)

5. Validity of Conclusions and Improvements

We believe that overall, our measurements and data were fairly accurate. However, the main issue we encountered (the bottleneck in our process) was the image recognition algorithm used to detect the position of the balls. At times, the program struggled to detect the balls, possibly due to slight changes in lighting or other environmental factors. This led to gaps in our data, where no position was recorded at specific moments. Using a camera to determine the balls' positions also introduces inherent limitations, such as lens distortion and the restricted width of the camera's frame; in some recordings, the red ball moved out of frame quickly. Despite these issues, we believe the automated ball release mechanism was accurate and reliable enough not to be the limiting factor in our measurements.

In future iterations of this experiment, we aim to improve the accuracy of our measurements by using a more precise method for tracking the balls' positions, such as infrared markers and sensors. This would also enable us to track the balls' movements in all three axes, rather than just two, and allow for more accurate measurement of rotational velocity. Additionally, the experiment could be extended by exploring a wider range of drop heights or by varying the mass of either ball.

6. Acknowledgements

We would like to thank our Head of Physics, Mr. Turner, for his ongoing support throughout our research, and for providing us with the laboratory space to conduct our experiments. We are also grateful to Sutton Grammar School for offering the necessary resources and equipment, which were essential to the success of this project.

7. Raw Data

The raw data collected from the experiments, including all .mp3 video recordings, velocity readings, height measurements, and other relevant details, are available for review in our public GitHub repository. This includes all code and software written for data analysis and graphing.

Please visit the following link to access the full dataset:

<https://github.com/BiancaIvanova/BPh0-Experimental-Challenge-2025>

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